Cancer Cell, Volume *39*

Supplemental information

Single-cell analyses reveal key immune cell subsets associated with response to PD-L1 blockade in triple-negative breast cancer

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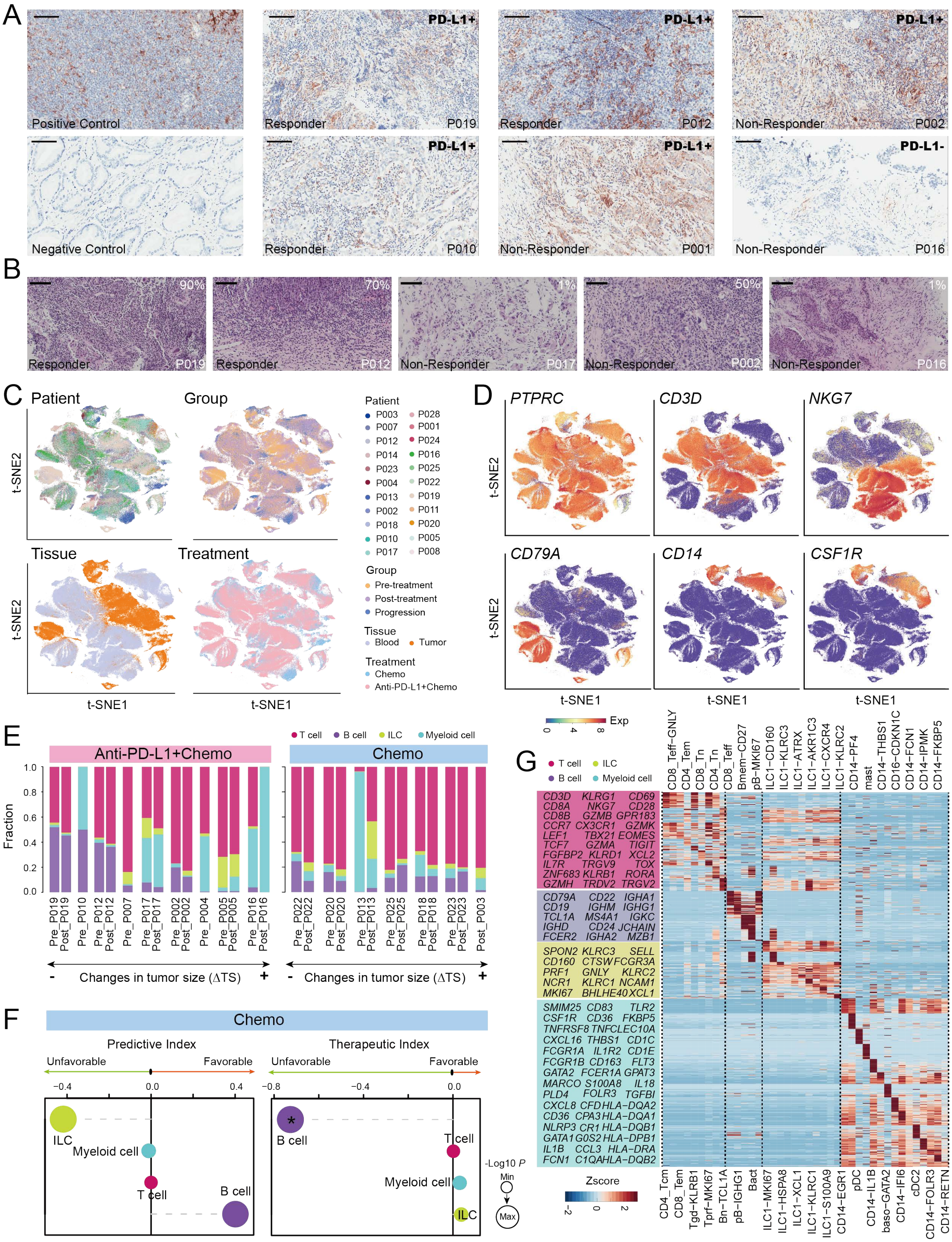

A
PD-L1+
Positive Control
Responder P019
Responder P012
Non-Responder P002
Negative Control
Responder P010
Non-Responder P001
PD-L1-
Non-Responder P016
B
90% Responder P019
70% Responder P012
1% Non-Responder P017
50% Non-Responder P002
1% Non-Responder P016
C
Patient
Group
Tissue
Treatment
t-SNE2
t-SNE1
Patient
P003 P028
P007 P001
P012 P024
P014 P016
P023 P025
P004 P022
P013 P019
P002 P011
P018 P020
P010 P005
P017 P008
Group
Pre-treatment
Post-treatment
Progression
Tissue
Blood Tumor
Treatment
Chemo
Anti-PD-L1+Chemo
D
PTPRC
CD3D
NKG7
CD79A
CD14
CSF1R
t-SNE2
t-SNE1
Exp
0 4 8
E
T cell B cell ILC Myeloid cell
Anti-PD-L1+Chemo
Chemo
Fraction
1.0 0.8 0.6 0.4 0.2 0.0
Pre_P019 Post_P019 Pre_P010 Pre_P012 Post_P012 Pre_P007 Pre_P017 Post_P017 Pre_P002 Post_P002 Pre_P004 Pre_P005 Post_P005 Pre_P016 Post_P016
Pre_P022 Post_P022 Pre_P020 Post_P020 Pre_P013 Post_P013 Pre_P025 Post_P025 Pre_P018 Post_P018 Pre_P023 Post_P023 Post_P003
- Changes in tumor size (ΔTS) +
- Changes in tumor size (ΔTS) +
F
Chemo
Predictive Index
Unfavorable Favorable
−0.4 0.0 0.4
ILC
Myeloid cell
T cell
B cell
Therapeutic Index
Unfavorable Favorable
−0.8 −0.4 0.0
*
B cell
T cell
Myeloid cell
ILC
-Log10 P
Min
Max
G
T cell ILC
B cell Myeloid cell
CD3D KLRG1 CD69
CD8A NKG7 CD28
CD8B GZMB GPR183
CCR7 CX3CR1 GZMK
LEF1 TBX21 EOMES
TCF7 GZMA TIGIT
FGFBP2 KLRD1 XCL2
IL7R TRGV9 TOX
ZNF683 KLRB1 RORA
GZMH TRDV2 TRGV2
CD79A CD22 IGHA1
CD19 IGHM IGHG1
TCL1A MS4A1 IGKC
IGHD CD24 JCHAIN
FCER2 IGHA2 MZB1
SPON2 KLRC3 SELL
CD160 CTSW FCGR3A
PRF1 GNLY KLRC2
NCR1 KLRC1 NCAM1
MKI67 BHLHE40 XCL1
SMIM25 CD83 TLR2
CSF1R CD36 FKBP5
TNFRSF8 TNF CLEC10A
CXCL16 THBS1 CD1C
FCGR1A IL1R2 CD1E
FCGR1B CD163 FLT3
GATA2 FCER1A GPAT3
MARCO S100A8 IL18
PLD4 FOLR3 TGFBI
CXCL8 CFD HLA-DQA2
CD36 CPA3 HLA-DQA1
NLRP3 CR1 HLA-DQB1
GATA1 G0S2 HLA-DPB1
IL1B CCL3 HLA-DRA
FCN1 C1QA HLA-DQB2
CD8_Teff-GNLY CD4_Tem CD8_Tn CD4_Tn CD8_Teff Bmem-CD27 pB-MKI67 ILC1-CD160 ILC1-KLRC3 ILC1-ATRX ILC1-AKR1C3 ILC1-CXCR4 ILC1-KLRC2 CD14-PF4 mast CD14-THBS1 CD16-CDKN1C CD14-FCN1 CD14-IPMK CD14-FKBP5
CD4_Tcm CD8_Tem Tgd-KLRB1 Tprf-MKI67 Bn-TCL1A pB-IGHG1 Bact ILC1-MKI67 ILC1-HSPA8 ILC1-XCL1 ILC1-KLRC1 ILC1-S100A9 CD14-EGR1 pDC CD14-IL1B baso-GATA2 CD14-IFI6 cDC2 CD14-FOLR3 CD14-RETN
Zscore
-2 0 2

Supplementary Figure 1. The Immune Cell Atlas in TNBC Patients, related to Figure 1.

(A) IHC staining of PD-L1 protein expression in representative tumors. Positive control, tonsil; Negative control, prostate. Original magnification, x20; Scale bar, 100 μm.

(B) H&E staining in representative tumors. Original magnification, x20; Scale bar, 100 μm.

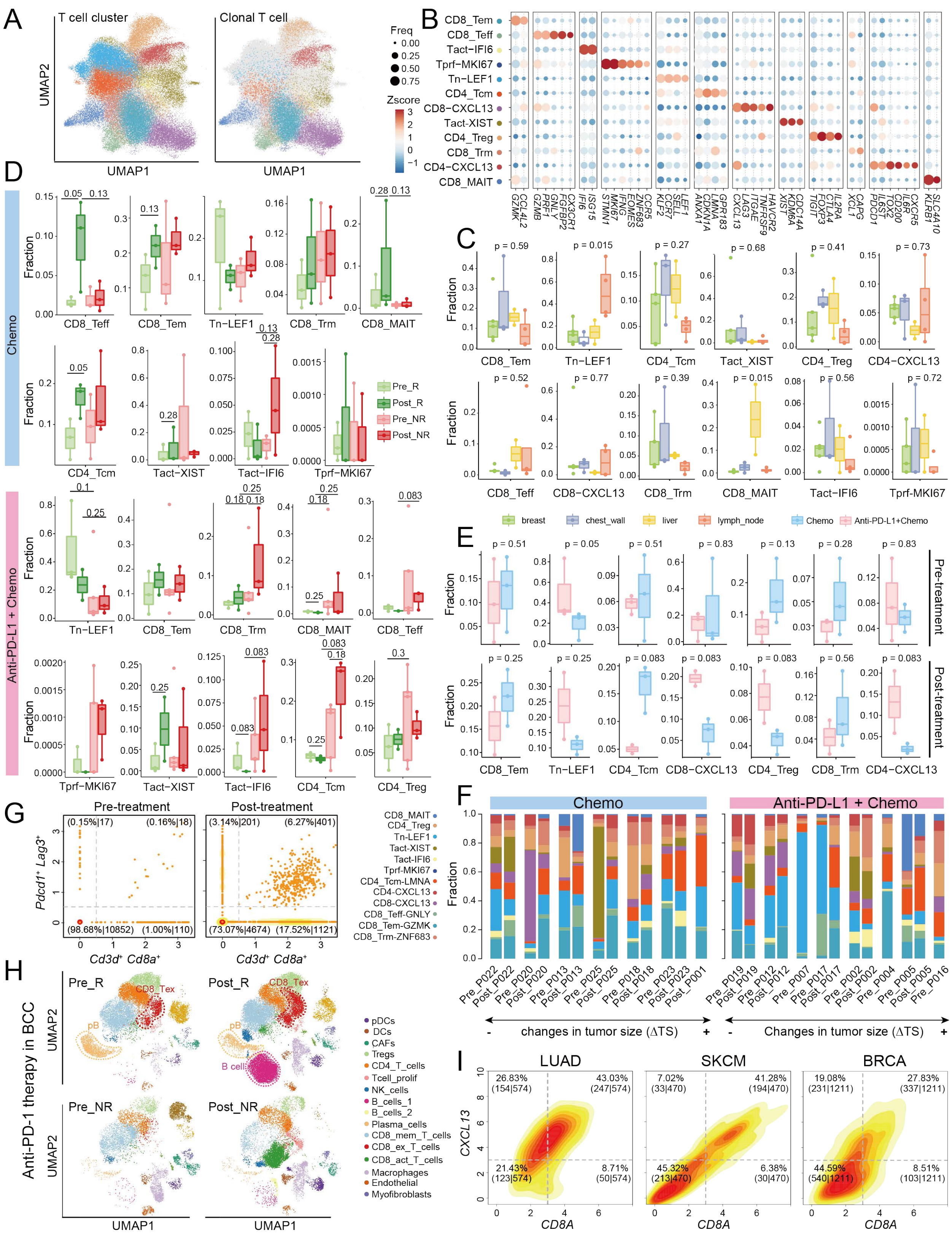
(C) t-SNE plots showing the origins of individual immune cells in TNBC patients.

(D) t-SNE plots showing the expressions of selected marker genes in major immune cell types.

(E) The cellular compositions of major immune cell types in TNBC tumors pre- and post-treatment.

(F) The predictive indices and therapeutic indices of major immune cell types in TNBC tumors treated with paclitaxel. Dot size represents the significance evaluated by -Log10 (*P*-value). *, 0.01 < *P*-value < 0.05.

(G) The expression patterns of signature genes in distinct blood-derived immune cell clusters.



A
T cell cluster
Clonal T cell
UMAP1
UMAP2
Freq
0.00
0.25
0.50
0.75
Zscore
B
CD8_Tem
CD8_Teff
Tact−IFI6
Tprf−MKI67
Tn−LEF1
CD4_Tcm
CD8−CXCL13
Tact−XIST
CD4_Treg
CD8_Trm
CD4−CXCL13
CD8_MAIT
C
Fraction
p = 0.59
p = 0.015
p = 0.27
p = 0.68
p = 0.41
p = 0.73
p = 0.52
p = 0.77
p = 0.39
p = 0.015
p = 0.56
p = 0.72
breast
chest_wall
liver
lymph_node
Chemo
Anti-PD-L1+Chemo
D
Chemo
Anti-PD-L1 + Chemo
Pre_R
Post_R
Pre_NR
Post_NR
E
Pre-treatment
Post-treatment
F
Chemo
Anti-PD-L1 + Chemo
changes in tumor size (ΔTS)
G
Pre-treatment
Post-treatment
Pdcd1+ Lag3+
Cd3d+ Cd8a+
(0.15%|17)
(0.16%|18)
(98.68%|10852)
(1.00%|110)
(3.14%|201)
(6.27%|401)
(73.07%|4674)
(17.52%|1121)
CD8_MAIT
CD4_Treg
Tn-LEF1
Tact-XIST
Tact-IFI6
Tprf-MKI67
CD4_Tcm-LMNA
CD4-CXCL13
CD8-CXCL13
CD8_Teff-GNLY
CD8_Tem-GZMK
CD8_Trm-ZNF683
H
Anti-PD-1 therapy in BCC
Pre_R
Post_R
Pre_NR
Post_NR
CD8_Tex
pB
B cells
pDCs
DCs
CAFs
Tregs
CD4_T_cells
Tcell_prolif
NK_cells
B_cells_1
B_cells_2
Plasma_cells
CD8_mem_T_cells
CD8_ex_T_cells
CD8_act_T_cells
Macrophages
Endothelial
Myofibroblasts
I
LUAD
SKCM
BRCA
CXCL13
CD8A
26.83% (154|574)
43.03% (247|574)
21.43% (123|574)
8.71% (50|574)
7.02% (33|470)
41.28% (194|470)
45.32% (213|470)
6.38% (30|470)
19.08% (231|1211)
27.83% (337|1211)
44.59% (540|1211)
8.51% (103|1211)

Supplementary Figure 2. Tumor-Infiltrating T Cell Subsets and Their Temporal Dynamics, related to Figure 2.

(A) UMAP plots showing T cell clusters and their clonal expansions in TNBC tumors. Clonal T cells were defined as those with TCR sequences harbored by at least two cells.

(B) The expression patterns of selected marker genes in distinct tumor-infiltrating T cell clusters.

(C) Boxplots showing the tissue origins of distinct tumor-infiltrating T cell clusters. One-way ANOVA test.

(D) Boxplots showing the temporal dynamics of distinct tumor-infiltrating T cell clusters. Black lines with different lengths indicate which two groups were compared. Kruskal-Wallis test.

(E) Boxplots showing the cellular compositions of distinct T cell clusters in responsive tumors treated with different regimens. Kruskal-Wallis test.

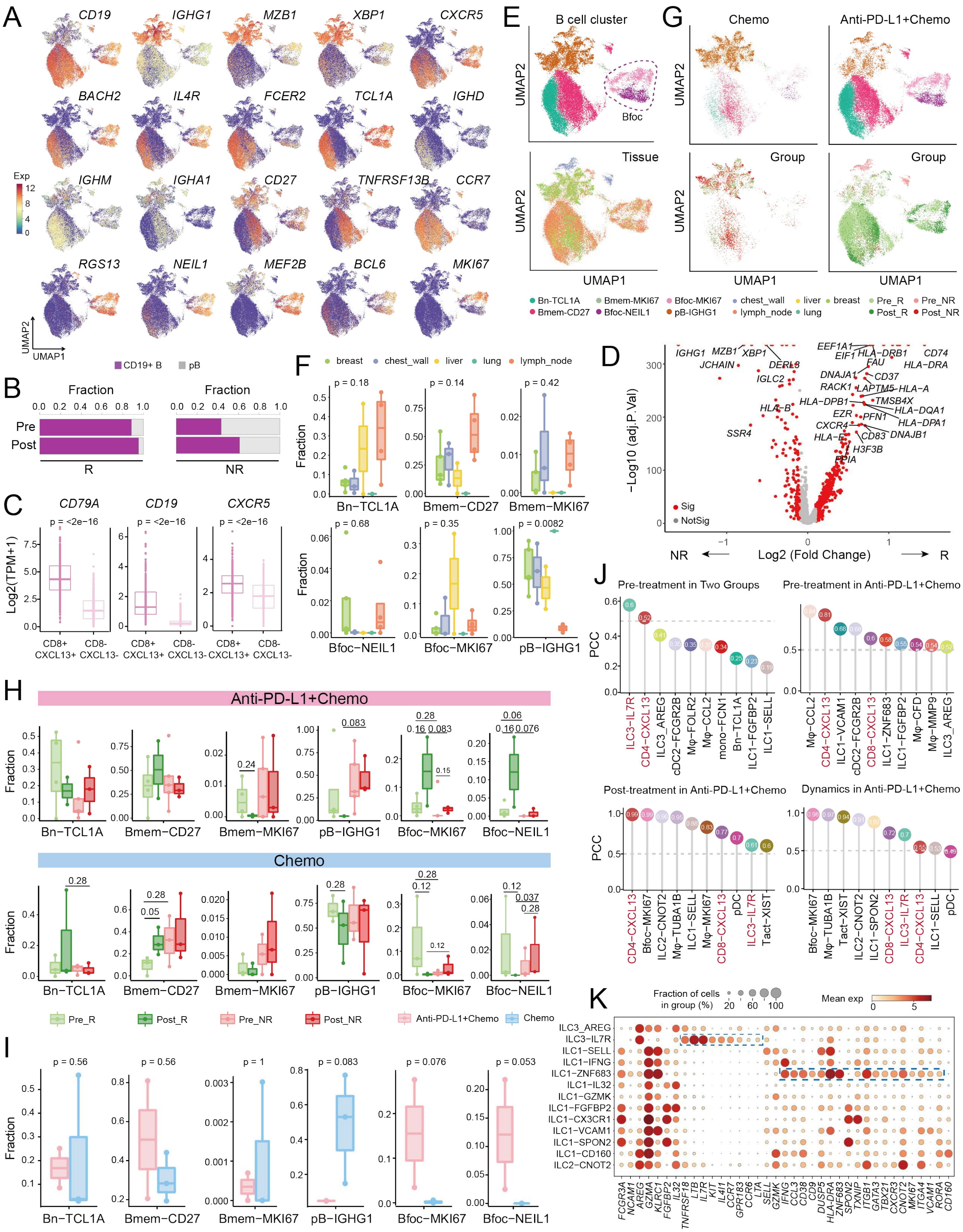
(F) Barplots showing the cellular compositions of distinct tumor-infiltrating T cell clusters in individual TNBC patients. Patients were ranked based on their tumor size changes.

(G) *In silico* flow cytometry (FACS) plots showing the alteration of *Cd3d*$^{+}$ *Cd8a*$^{+}$ *Pdcd1*$^{+}$ *Lag3*$^{+}$ T cells in T11-Apobec murine model of BC following ICB treatment (Hollern et al., 2019). The geometric mean of expression levels of *Cd3d* and *Cd8a*, and those of *Pdcd1* and *Lag3* were used to evaluate the percentages of *Cd3d*$^{+}$ *Cd8a*$^{+}$ *Pdcd1*$^{+}$ *Lag3*$^{+}$ T cells.

(H) UMAP plots showing the alterations of tumor-infiltrating immune cell subtypes in BCC patients pre- and post-treatment with ICB (Yost et al., 2019).

(I) Density plots showing the proportions of tumor samples that express high levels of *CD8A* and *CXCL13* in TCGA lung adenocarcinoma (LUAD), skin cutaneous melanoma (SKCM) and breast invasive carcinoma (BRCA). Expression level was measured by Log2 (TPM+1).

For all box plots: box middle lines, median; box limits, upper and lower quartiles; box whiskers, 1.5 × the interquartile range.


A
CD19
IGHG1
MZB1
XBP1
CXCR5
BACH2
IL4R
FCER2
TCL1A
IGHD
IGHM
IGHA1
CD27
TNFRSF13B
CCR7
RGS13
NEIL1
MEF2B
BCL6
MKI67
Exp
UMAP2
UMAP1
B
CD19+ B
pB
Fraction
Pre
Post
R
NR
C
CD79A
CD19
CXCR5
p = <2e−16
Log2(TPM+1)
CD8+ CXCL13+
CD8- CXCL13-
E
B cell cluster
Bfoc
Tissue
UMAP2
UMAP1
G
Chemo
Anti-PD-L1+Chemo
Group
Bn-TCL1A
Bmem-CD27
Bmem-MKI67
Bfoc-NEIL1
Bfoc-MKI67
pB-IGHG1
chest_wall
lymph_node
liver
lung
breast
Pre_R
Post_R
Pre_NR
Post_NR
F
Fraction
p = 0.18
p = 0.14
p = 0.42
p = 0.68
p = 0.35
p = 0.0082
D
−Log10 (adj.P.Val)
Log2 (Fold Change)
NR
R
Sig
NotSig
J
Pre-treatment in Two Groups
Pre-treatment in Anti-PD-L1+Chemo
Post-treatment in Anti-PD-L1+Chemo
Dynamics in Anti-PD-L1+Chemo
PCC
H
Anti-PD-L1+Chemo
Chemo
Fraction
I
p = 0.56
p = 0.56
p = 1
p = 0.083
p = 0.076
p = 0.053
K
Fraction of cells in group (%)
Mean exp

Supplementary Figure 3. Features of Tumor-infiltrating B Cell Clusters, related to Figure 3.

(A) UMAP plots showing the expression patterns of selected genes in tumor-infiltrating B cells.

(B) Barplots showing the compositional changes of *CD19*$^{+}$ B and pB cells in responders (R) and non-responders (NR) of the combination therapy group.

(C) Boxplots showing the expression levels of selected genes in TCGA BRCA patients grouped by *CD8A* and *CXCL13* expressions in Figure S2I.

(D) The volcano plot showing DEGs in B cells enriched in pre-treatment responsive and non-responsive tumors of the combination therapy group. Sig, significant; BH adjusted P-value < 0.01 and Log2 (fold change) ≥ 0.1 or Log2 (fold change) ≤ -0.1. Limma moderated t-test.

(E) UMAP plots showing the distributions and tissue origins of tumor-infiltrating B cell clusters.

(F) Boxplots showing the tissue origins of distinct B cell clusters. One-way ANOVA test.

(G) UMAP plots showing the distributions of B cell clusters in different treatment groups.

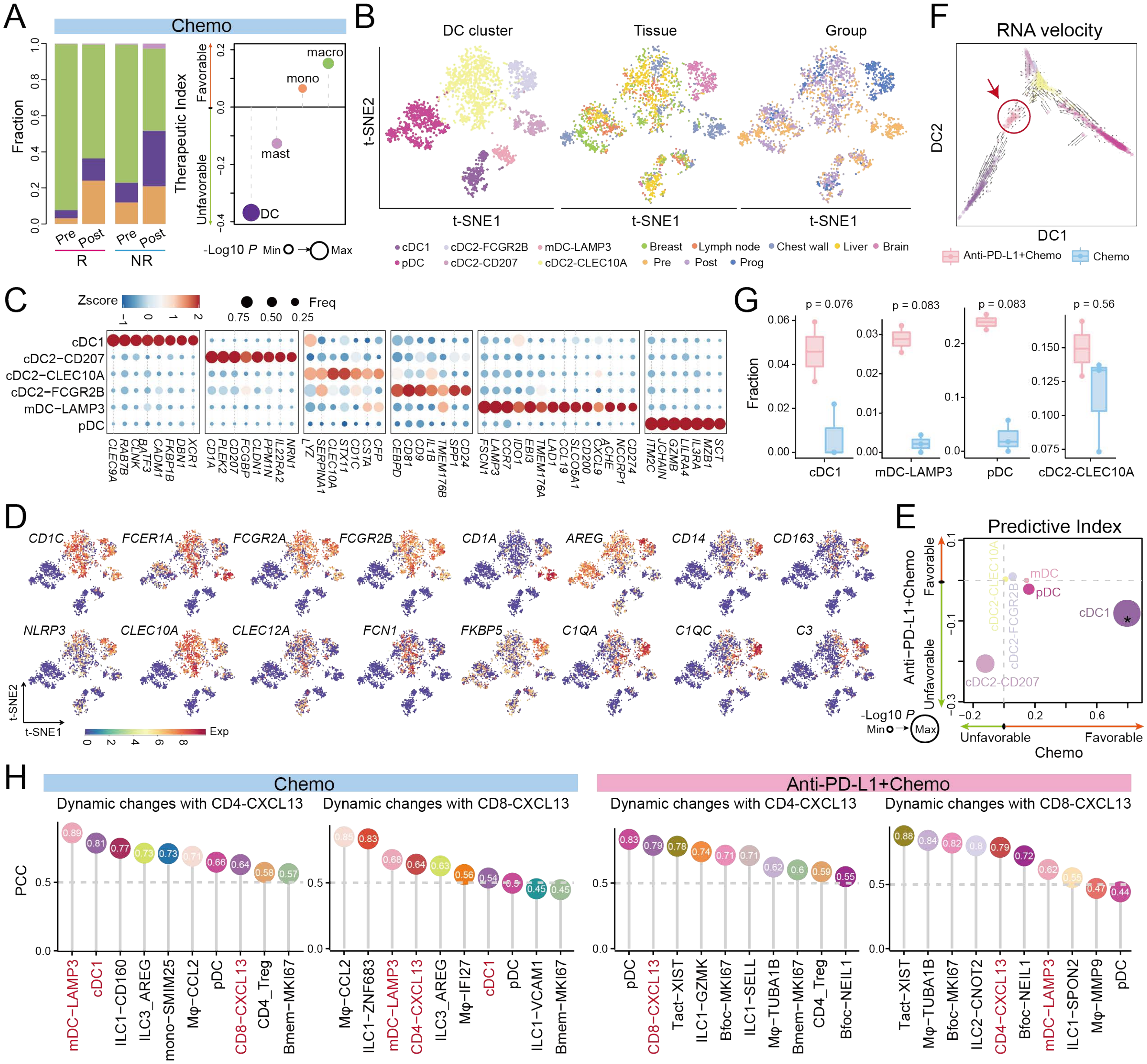
(H) Boxplots showing the temporal dynamics of distinct B cell clusters. Black lines with different lengths indicate which two groups were compared. Kruskal-Wallis test.

(I) Boxplots showing the cellular compositions of distinct B cell clusters in responsive patients following different treatment regimens. Kruskal-Wallis test.

(J) Correlations of distinct tumor-infiltrating immune cells with Bfoc in different treatment groups. PCC, Pearson correlation coefficient.

(K) The expression patterns of selected marker genes in distinct tumor-infiltrating ILC clusters.

For all box plots: box middle lines, median; box limits, upper and lower quartiles; box whiskers, 1.5 × the interquartile range.



A
Chemo
Fraction
Pre Post Pre Post
R NR
Therapeutic Index
Favorable
Unfavorable
macro
mono
mast
DC
-Log10 P Min Max
B
DC cluster
Tissue
Group
t-SNE2
t-SNE1
cDC1 cDC2-FCGR2B mDC-LAMP3 pDC cDC2-CD207 cDC2-CLEC10A
Breast Lymph node Chest wall Liver Brain
Pre Post Prog
F
RNA velocity
DC2
DC1
Anti-PD-L1+Chemo Chemo
C
Zscore
Freq
cDC1
cDC2−CD207
cDC2−CLEC10A
cDC2−FCGR2B
mDC−LAMP3
pDC
G
p = 0.076
p = 0.083
p = 0.083
p = 0.56
Fraction
cDC1
mDC-LAMP3
pDC
cDC2-CLEC10A
D
CD1C FCER1A FCGR2A FCGR2B CD1A AREG CD14 CD163
NLRP3 CLEC10A CLEC12A FCN1 FKBP5 C1QA C1QC C3
t-SNE2
t-SNE1
Exp
E
Predictive Index
Anti-PD-L1+Chemo
Favorable
Unfavorable
mDC
pDC
cDC1
cDC2-CLEC10A
cDC2-FCGR2B
cDC2-CD207
-Log10 P
Min Max
Unfavorable Favorable
Chemo
H
Chemo
Anti-PD-L1+Chemo
Dynamic changes with CD4-CXCL13
Dynamic changes with CD8-CXCL13
PCC
mDC−LAMP3 cDC1 ILC1−CD160 ILC3_AREG mono−SMIM25 Mφ−CCL2 pDC CD8−CXCL13 CD4_Treg Bmem−MKI67
Mφ−CCL2 ILC1−ZNF683 mDC−LAMP3 CD4−CXCL13 ILC3_AREG Mφ−IFI27 cDC1 pDC ILC1−VCAM1 Bmem−MKI67
pDC CD8−CXCL13 Tact−XIST ILC1−GZMK Bfoc−MKI67 ILC1−SELL Mφ−TUBA1B Bmem−MKI67 CD4_Treg Bfoc−NEIL1
Tact−XIST Mφ−TUBA1B Bfoc−MKI67 ILC2−CNOT2 CD4−CXCL13 Bfoc−NEIL1 mDC−LAMP3 ILC1−SPON2 Mφ−MMP9 pDC

Supplementary Figure 4. Characteristics of Tumor-infiltrating DC Subsets, related to Figure 4.

(A) The cellular compositions and dynamics of major myeloid subpopulations in TNBC tumors treated with paclitaxel.

(B) t-SNE plots showing the distributions of DC subsets.

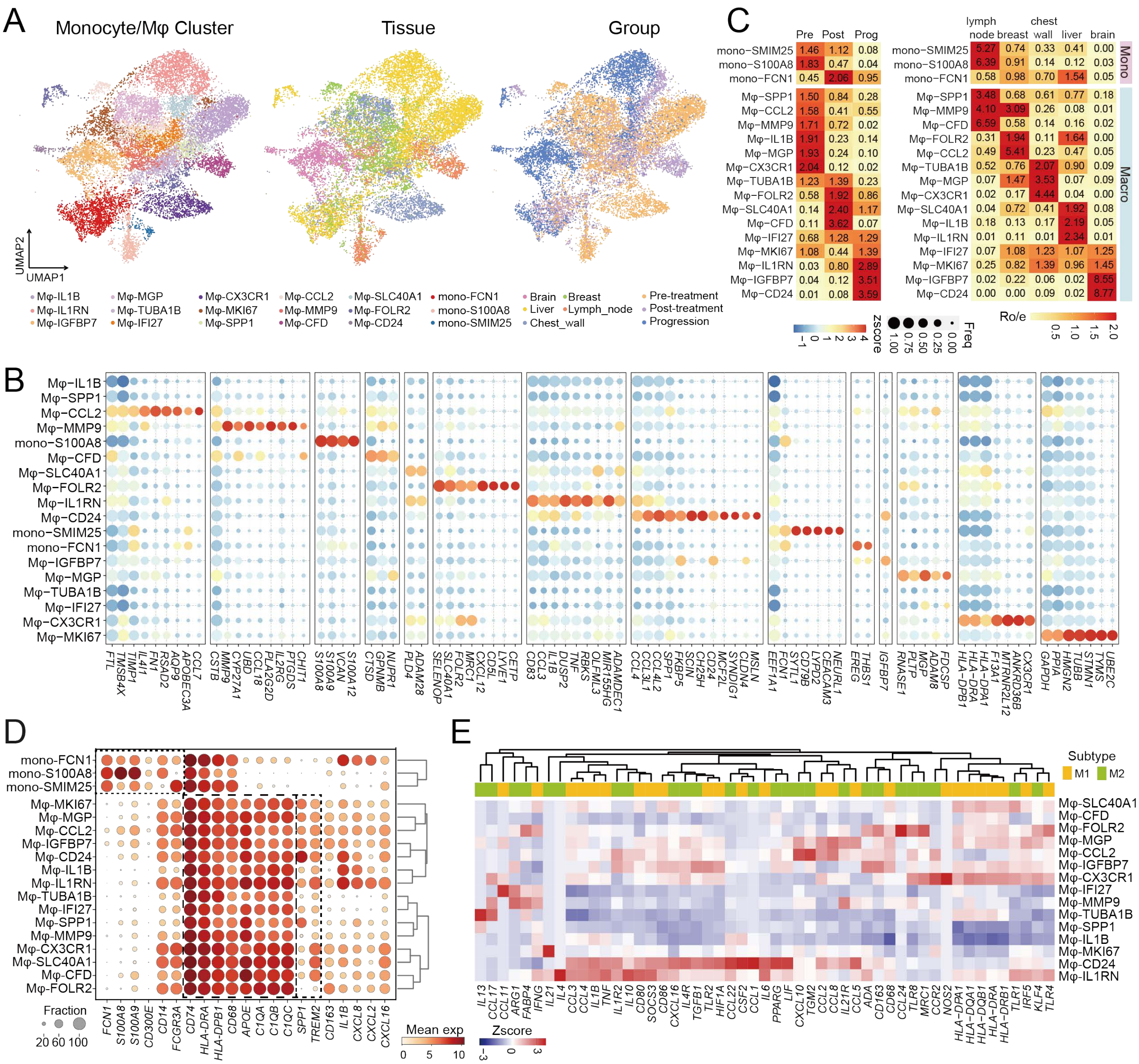
(C) Bubble heatmap and (D) t-SNE plots showing the expression patterns of selected marker genes in distinct DC clusters.

(E) The predictive indices of distinct DC subsets. Dot size represents the geometric mean of -Log10 (*P*-value) in different treatment groups.* bottom, 0.01< *P*-value < 0.05 for chemotherapy.

(F) The developmental trajectory of distinct DC clusters inferred by RNA velocity. DC, diffusion component.

(G) Boxplots showing the cellular compositions of distinct DC clusters in responsive patients following different treatment regimens. Box middle lines, median; box limits, upper and lower quartiles; box whiskers, 1.5 × the interquartile range. Kruskal-Wallis test.

(H) Correlations of distinct tumor-infiltrating immune cell subsets with CD4-CXCL13 and CD8-CXCL13 in their cellular proportion dynamics in different treatment groups. PCC, Pearson correlation coefficient.


A
Monocyte/Mφ Cluster
Tissue
Group
UMAP2
UMAP1
Mφ-IL1B
Mφ-IL1RN
Mφ-IGFBP7
Mφ-MGP
Mφ-TUBA1B
Mφ-IFI27
Mφ-CX3CR1
Mφ-MKI67
Mφ-SPP1
Mφ-CCL2
Mφ-MMP9
Mφ-CFD
Mφ-SLC40A1
Mφ-FOLR2
Mφ-CD24
mono-FCN1
mono-S100A8
mono-SMIM25
Brain
Breast
Liver
Lymph_node
Chest_wall
Pre-treatment
Post-treatment
Progression
B
C
Pre
Post
Prog
lymph node
breast
chest wall
liver
brain
Mono
Macro
Zscore
Freq
Ro/e
D
Fraction
Mean exp
E
Subtype
M1
M2
Zscore

Supplementary Figure 5. The Heterogeneity of Monocycles/Macrophages in TNBC Tumors, related to Figure 5.

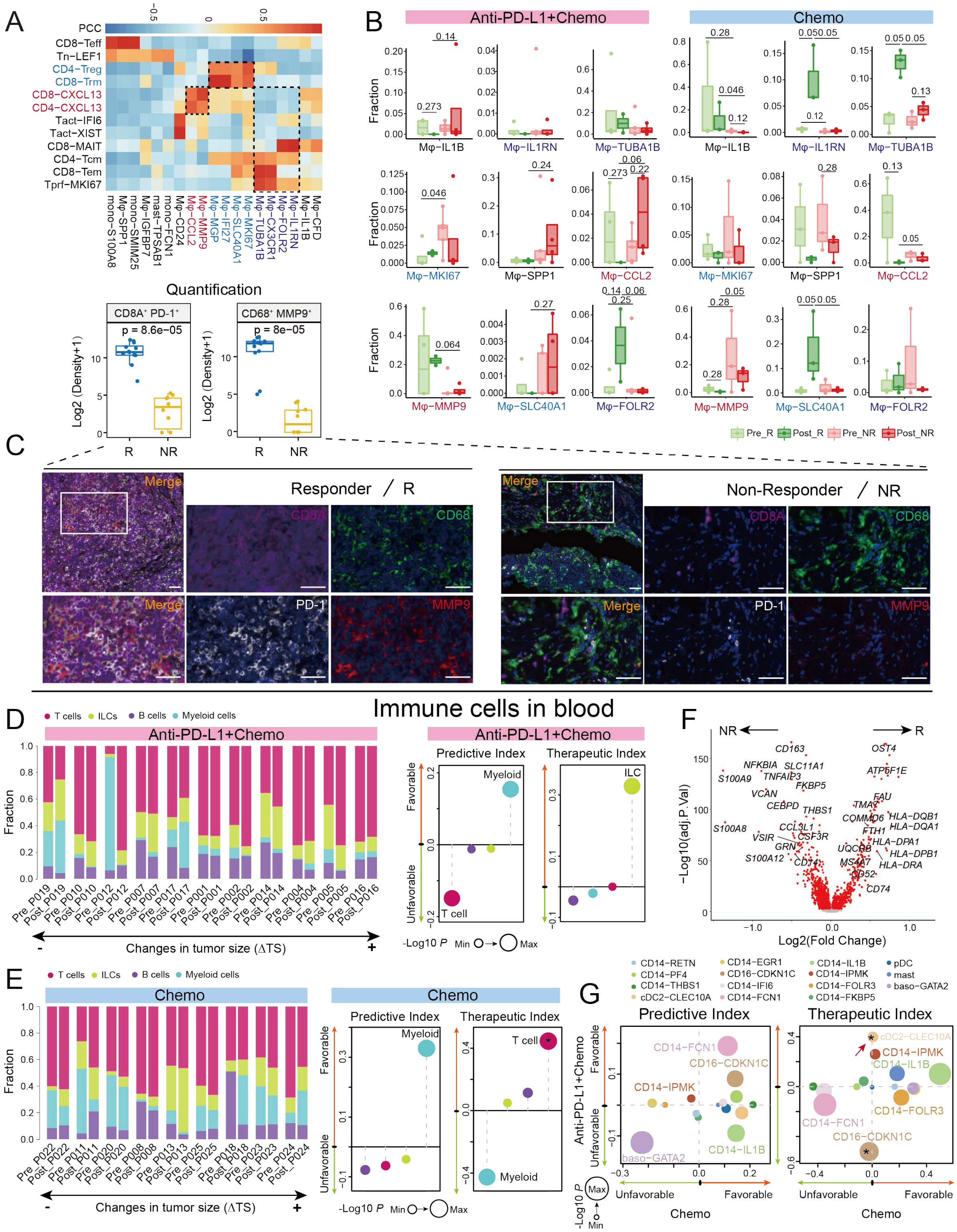
(A) UMAP plots showing the monocyte/macrophage cell clusters in TNBC tumors and their distributions in different tissues and groups.

(B) The expression patterns of signature genes of distinct monocyte/macrophage cell clusters.

(C) The enrichment of distinct monocyte/macrophage clusters in different groups and tissues (STAR Methods).

(D) Bubble heatmap showing the expressions of selected genes.

(E) Heatmap showing the expressions of canonical M1 or M2 phenotype-related genes.


A
PCC
-0.5
0
0.5
CD8-Teff
Tn-LEF1
CD4-Treg
CD8-Trm
CD8-CXCL13
CD4-CXCL13
Tact-IFI6
Tact-XIST
CD8-MAIT
CD4-Tcm
CD8-Tem
Tprf-MKI67
mono-S100A8
Mφ-SPP1
mono-SMIM25
Mφ-IGFBP7
mast-TPSAB1
mono-FCN1
Mφ-CD24
Mφ-CCL2
Mφ-MMP9
Mφ-MGP
Mφ-IFI27
Mφ-SLC40A1
Mφ-MKI67
Mφ-TUBA1B
Mφ-CX3CR1
Mφ-FOLR2
Mφ-IL1RN
Mφ-IL1B
Mφ-CFD
Quantification
CD8A+ PD-1+
p = 8.6e-05
CD68+ MMP9+
p = 8e-05
Log2 (Density+1)
R
NR
B
Anti-PD-L1+Chemo
Chemo
Fraction
Mφ-IL1B
Mφ-IL1RN
Mφ-TUBA1B
Mφ-MKI67
Mφ-SPP1
Mφ-CCL2
Mφ-MMP9
Mφ-SLC40A1
Mφ-FOLR2
Pre_R
Post_R
Pre_NR
Post_NR
C
Merge
Responder / R
CD8A
CD68
PD-1
MMP9
Non-Responder / NR
D
Immune cells in blood
T cells
ILCs
B cells
Myeloid cells
Anti-PD-L1+Chemo
Fraction
Changes in tumor size (ΔTS)
Predictive Index
Therapeutic Index
Favorable
Unfavorable
Myeloid
T cell
ILC
-Log10 P Min → Max
E
Chemo
T cell
Myeloid
F
NR
R
-Log10(adj.P.Val)
Log2(Fold Change)
CD163
OST4
NFKBIA
SLC11A1
S100A9
TNFAIP3
ATP5F1E
FKBP5
VCAN
FAU
CEBPD
THBS1
TMA7
COMMD6
HLA-DQB1
S100A8
CCL3L1
FTH1
HLA-DQA1
VSIR
CSF3R
HLA-DPA1
GRN
UQCRB
S100A12
CD14
HLA-DPB1
MS4A7
HLA-DRA
CD52
CD74
G
CD14-RETN
CD14-PF4
CD14-THBS1
cDC2-CLEC10A
CD14-EGR1
CD16-CDKN1C
CD14-IFI6
CD14-FCN1
CD14-IL1B
CD14-IPMK
CD14-FOLR3
CD14-FKBP5
pDC
mast
baso-GATA2
Predictive Index
Therapeutic Index
Anti-PD-L1+Chemo
Favorable
Unfavorable
Chemo
-Log10 P
Max
Min

Supplementary Figure 6. Features and Dynamics of Monocycles/Macrophages in TNBC Patients, related to Figure 5.

(A) Heatmap showing correlations of distinct monocycle/macrophage clusters with T cell clusters in their baseline cellular proportions in TNBC patients treated with paclitaxel plus atezolizumab. PCC, Pearson correlation coefficient.

(B) Boxplots showing the temporal dynamics of distinct macrophage subsets in TNBC tumors. Black lines with different lengths indicate which two groups were compared. Kruskal-Wallis test.

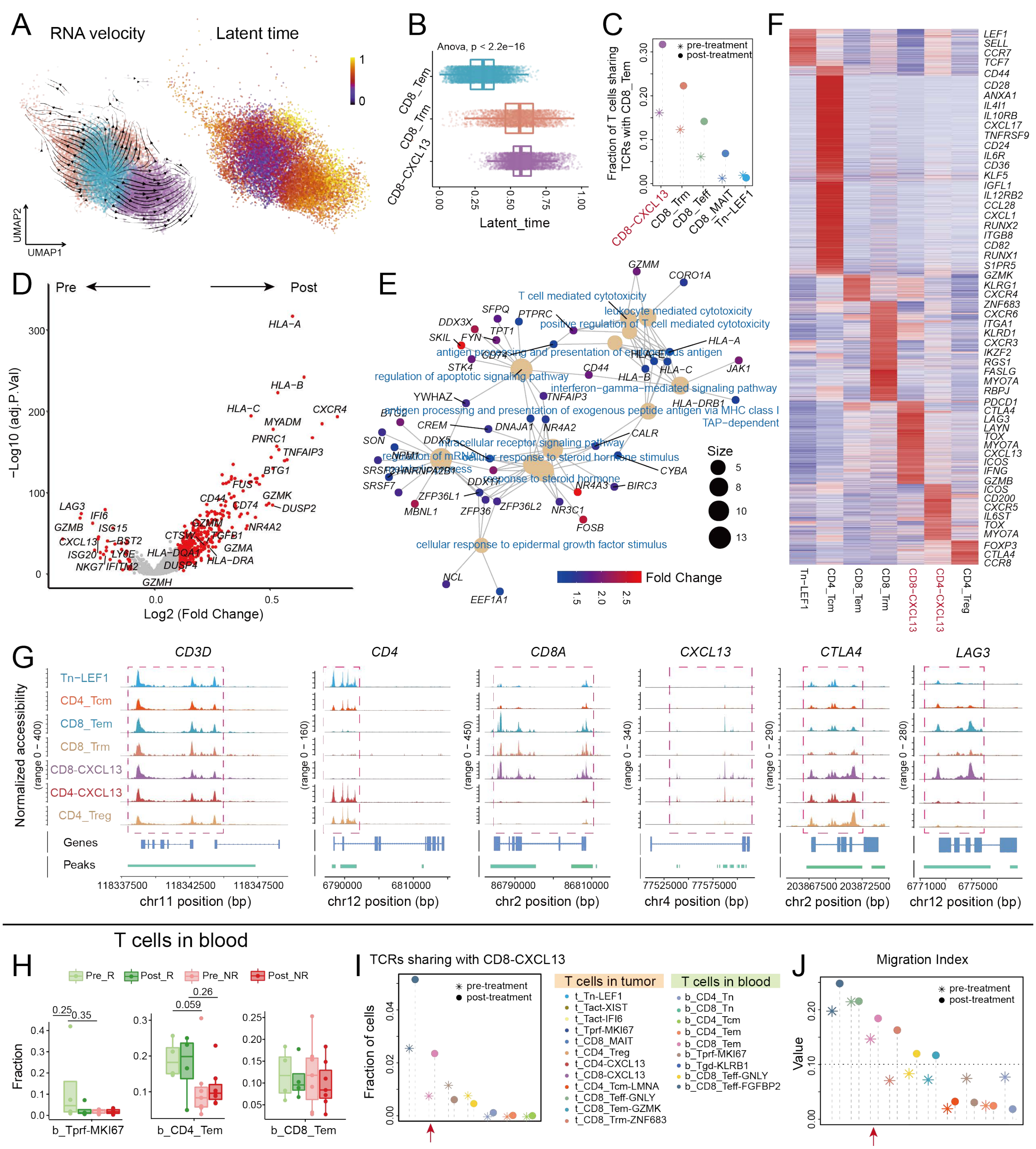
(C) Multi-color IHC staining in one representative responder (P019) and one non-responder (P016) of the combination therapy group. Scale bar, 30 μm. Each dot in boxplot represents one view. Wilcoxon test.

(D) The cellular compositions, predictive indices and therapeutic indices of blood-derived immune cells in TNBC patients treated with the combination therapy or (E) the chemotherapy. *, $0.01 < P\text{-value} < 0.05$.

(F) The volcano plot showing DEGs of baseline blood monocytes in responders and non-responders of the combination therapy group. BH adjusted P-value < 0.01. Limma moderated t-test.

(G) The predictive indices and therapeutic indices of distinct blood-derived myeloid subsets. * left, $0.01 < P\text{-value} < 0.05$ for the combination therapy group.

For all box plots: box middle lines, median; box limits, upper and lower quartiles; box whiskers, $1.5\times$ the interquartile range.


A
RNA velocity
Latent time
UMAP2
UMAP1
B
Anova, p < 2.2e−16
CD8_Tem
CD8_Trm
CD8−CXCL13
Latent_time
C
pre-treatment
post-treatment
Fraction of T cells sharing TCRs with CD8_Tem
CD8−CXCL13
CD8_Trm
CD8_Teff
CD8_MAIT
Tn−LEF1
D
Pre
Post
−Log10 (adj.P.Val)
Log2 (Fold Change)
HLA−A
HLA−B
HLA−C
CXCR4
MYADM
PNRC1
TNFAIP3
BTG1
FUS
CD44
CD74
GZMK
DUSP2
NR4A2
LAG3
IFI6
GZMB
ISG15
CXCL13
BST2
ISG20
LY6E
NKG7
IFITM2
CTSW
GZMM
TGFB1
HLA−DQA1
GZMA
DUSP4
HLA−DRA
GZMH
E
T cell mediated cytotoxicity
leukocyte mediated cytotoxicity
positive regulation of T cell mediated cytotoxicity
antigen processing and presentation of endogenous antigen
regulation of apoptotic signaling pathway
interferon−gamma−mediated signaling pathway
antigen processing and presentation of exogenous peptide antigen via MHC class I TAP−dependent
intracellular receptor signaling pathway
cellular response to steroid hormone stimulus
response to steroid hormone
cellular response to epidermal growth factor stimulus
Size
Fold Change
F
LEF1
SELL
CCR7
TCF7
CD44
CD28
ANXA1
IL10RB
CXCL17
TNFRSF9
CD24
IL6R
CD36
KLF5
IGFL1
IL12RB2
CCL28
CXCL1
RUNX2
ITGB8
CD82
RUNX1
S1PR5
GZMK
KLRG1
CXCR4
ZNF683
CXCR6
ITGA1
KLRD1
CXCR3
IKZF2
RGS1
FASLG
MYO7A
RBPJ
PDCD1
CTLA4
LAYN
LAG3
TOX
MYO7A
CXCL13
ICOS
IFNG
GZMB
ICOS
CD200
CXCR5
IL6ST
TOX
MYO7A
FOXP3
CTLA4
CCR8
Tn−LEF1
CD4_Tcm
CD8_Tem
CD8_Trm
CD8−CXCL13
CD4−CXCL13
CD4_Treg
G
CD3D
CD4
CD8A
CXCL13
CTLA4
LAG3
Normalized accessibility
Tn−LEF1
CD4_Tcm
CD8_Tem
CD8_Trm
CD8-CXCL13
CD4-CXCL13
CD4_Treg
Genes
Peaks
chr11 position (bp)
chr12 position (bp)
chr2 position (bp)
chr4 position (bp)
chr2 position (bp)
chr12 position (bp)
T cells in blood
H
Pre_R
Post_R
Pre_NR
Post_NR
Fraction
b_Tprf-MKI67
b_CD4_Tem
b_CD8_Tem
I
TCRs sharing with CD8-CXCL13
Fraction of cells
T cells in tumor
t_Tn-LEF1
t_Tact-XIST
t_Tact-IFI6
t_Tprf-MKI67
t_CD8_MAIT
t_CD4_Treg
t_CD4-CXCL13
t_CD8-CXCL13
t_CD4_Tcm-LMNA
t_CD8_Teff-GNLY
t_CD8_Tem-GZMK
t_CD8_Trm-ZNF683
T cells in blood
b_CD4_Tn
b_CD8_Tn
b_CD4_Tcm
b_CD4_Tem
b_CD8_Tem
b_Tprf-MKI67
b_Tgd-KLRB1
b_CD8_Teff-GNLY
b_CD8_Teff-FGFBP2
J
Migration Index
Value
pre-treatment
post-treatment

Supplementary Figure 7. Transcriptomic and Epigenetic Features of CD8-CXCL13 T cells, related to Figure 6.

(A) The developmental trajectory of $CD8^+$ T cell subsets of Tem, Trm and CD8-CXCL13 inferred by RNA velocity.

(B) Boxplots showing the inferred latent times of $CD8^+$ T cell clusters based on their developmental trajectory.

(C) The percentages of clonal T cells that shared TCRs with CD8 Tem in distinct T cell subsets in responsive tumors pre- and post-treatment with paclitaxel plus atezolizumab.

(D) The volcano plot showing DEGs of CD8-CXCL13 in responsive patients pre- and post-treatment with the combination therapy. BH adjusted *P*-value < 0.01. Limma moderated t-test.

(E) The enriched pathways for genes highly expressed in CD8-CXCL13 in responsive patients following the combination therapy.

(F) Heatmap showing the signature peaks and associated genes in distinct T cell clusters.

(G) ATAC-seq tracks showing the chromatin accessibilities of *CD3D*, *CD4*, *CD8A*, *CXCL13*, *CTLA4* and *LAG3* in distinct tumor-infiltrating T cell clusters.

(H) Boxplots showing the temporal dynamics of selected blood-derived T cell subsets in TNBC patients treated with the combination therapy.

(I) The percentages of clonal T cells that shared TCRs with tumor CD8-CXCL13 in distinct T cell subsets in responsive patients pre- and post-treatment with the combination therapy.

(J) The STARTRAC migration index of blood CD8 Tem cells in responsive patients pre- and post-treatment with the combination therapy.

For all box plots: box middle lines, median; box limits, upper and lower quartiles; box whiskers, $1.5\times$ the interquartile range.